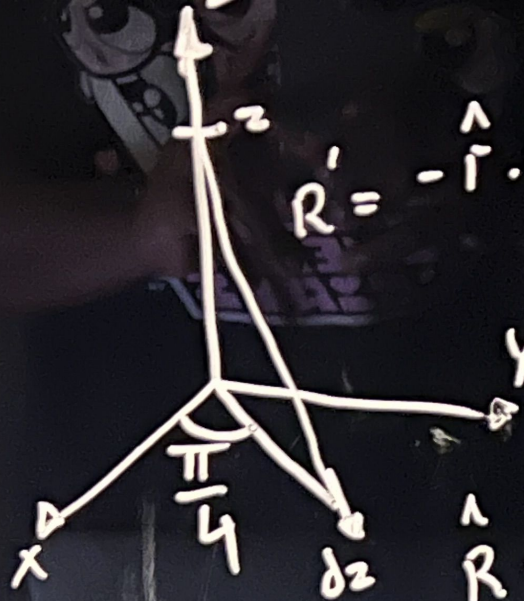
4.15



$$R' = -\hat{r}\,.02 + \hat{z}z$$

$$r = 2\text{ cm}$$

$$0 \le \phi \le \frac{\pi}{4}$$

$$R = \hat{x}\,0.02\cos\phi - \hat{Y}\,0.02\sin\phi + \hat{z}z$$

A)

$$dl = r\,d\phi$$

$$E = \frac{1}{4\pi\varepsilon_0}\int \frac{R\,P_1\,dl'}{R'^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\int_{\phi=0}^{\pi/4} \frac{P_1(-\hat{x}\,0.02\cos\phi - \hat{Y}\,0.02\sin\phi + \hat{z}z)\times 0.02}{(0.02^2 + z^2)^{3/2}}\,d\phi$$

$$\frac{5\times10^{-6}\times 0.02}{4\pi\varepsilon_0(0.02^2+z^2)^{3/2}}\int_{\phi=0}^{\pi/4}(-\hat{x}\,0.02\cos\phi - \hat{Y}\,0.02\sin\phi + \hat{z}z)\,d\phi$$

$$\frac{898.8}{((0.02)^2+z^2)^{3/2}} \times \left[-\hat{x}\,0.02\sin\phi - \hat{Y}\,0.02\cos\phi + \hat{z}z\phi\right]_{\phi=0}^{\pi/4}$$

$$\frac{898.8}{(0.02^2+z^2)}\left[-\hat{x}\,0.014 - \hat{Y}\,0.006 + \hat{z}\,0.78z\right]\frac{V}{m}$$

4.15

A) $dl = r\,d\phi$

$$E = \frac{1}{4\pi\varepsilon_0}\int \frac{R\,P_1\,dl}{R^{1/2}}$$

$$E = \frac{1}{4\pi\varepsilon_0}\int_{\phi=0}^{\pi/4} \frac{P_1(-\hat{x}\,0.02\cos\phi - \hat{y}\,0.02\sin\phi + \hat{z}z)\times 0.02}{(0.02^2+z^2)^{3/2}}\,d\phi$$

$$\frac{5\times10^{-6}\times0.02}{4\pi\varepsilon_0(0.02^2+z^2)^{3/2}}\int_{\phi=0}^{\pi/4}(-\hat{x}\,0.02\cos\phi - \hat{y}\,0.02\sin\phi + \hat{z}z)\,d\phi$$

$$\frac{898.8}{(0.02)^2+z^2)^{3/2}}\times\left[-\hat{x}\,0.02\sin\phi - \hat{y}\,0.02\cos\phi + \hat{z}z\phi\right]_{\phi=0}^{\pi/4}$$

$$\frac{898.8}{(0.02^2+z^2)}\left[-\hat{x}\,0.014 - \hat{y}\,0.006 + \hat{z}\,0.78z\right]\frac{V}{m}$$

4.15

B.) $z = 5\,cm$

$= -\hat{x}\,80.573 - \hat{y}\,34.531 + \hat{z}\,224.455\ \frac{nV}{m}$

C.) $= -\hat{x}\,80.573 - \hat{y}\,34.531 - \hat{z}\,224.455\ \frac{n}{}$

4.24

$a < R < b$

$$\oint E \cdot ds = \frac{Q_E}{\epsilon_0}$$

$$E = \frac{Q_E}{4\pi R^2 \epsilon_0}$$

$$E = \frac{Q_1}{4\pi R^2 \epsilon_0} \hat{R}$$

$R > b$

$$E = \frac{Q_1 + Q_2}{4\pi R^2 \epsilon_0} \hat{R}$$

$R < a$

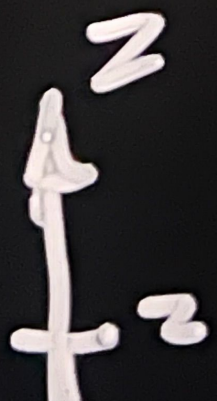
$E = 0$

$$\frac{Q_{enc}}{\epsilon_0}$$

$$\frac{Q_1}{\epsilon_0}$$

$$= \frac{Q_1}{\epsilon_0}$$

ϵ_0



4.26

$\rho_v = 5re^{-r}\left(\frac{C}{m^3}\right)$

$\nabla \cdot D = \rho_v$

$D = \hat{r} D$

$\nabla \cdot \hat{r} D_r = 5re^{-r}$

$\frac{\partial}{\partial r}(r^2 D_r) = 5r^2 e^{-r}$

$\int_0^r \frac{\partial}{\partial r}(r^2 D_r)\,dr = \int_0^r (5r^2 e^{-r})\,dr$

$D_r[r]_0^r = 5[2(1-e^{-r}(1+r)) - r^2 e^{-r}]$

$D = \hat{r}5\left[\frac{2}{r^2}(1-e^{-r}(1+r))\right] - re^{-r}$

4.27

a) $r < 1m$

$$\oint D \cdot ds = 0$$

$$D = 0$$

b) $1m < r < 3m$

$$\oint D \cdot ds = \int_V \rho_v \, dv$$

$$D_r \times 2\pi r L = \int_{r=1}^{r} \int_{\phi=0}^{2\pi} \int_{z=0}^{L} \rho_{v_0} r \, dr \, d\phi \, dz$$

$$D_r \times 2\pi r L = \rho_{v_0} \cdot \frac{r^2}{2} \left[r \cdot 2\pi \cdot z \right]_0^L$$

$$= \frac{\rho_{v_0} (r^2 - 1) 2\pi L}{2}$$

$$D = \frac{\rho_{v_0} (r^2 - 1)}{2r} a_r$$

c) $r > 3m$

$$D = \frac{4 \rho_{v_0} \cdot 4 \cdot 2\pi L}{r} a_r$$

4.35

A) $E = \frac{qd}{4\pi\varepsilon_0 R^3}\left(2\cos(\theta)\hat{R} + \sin(\theta)\hat{\theta}\right)$

$E = 4\frac{mV}{m}$

$4\frac{mV}{m} = \frac{q(1\,cm)}{4\pi\varepsilon_0 (1m)^3}\left(2\cos(0^\circ)\hat{R} + \sin(0^\circ)\hat{\theta}\right)$

$4\times10^{-3} = \frac{q(1\times10^{-2})}{4\pi\varepsilon_0}(2)$

$q = (0.8\pi\varepsilon_0)C \qquad d = 1\times10^{-2}\,m$

$R = 2m$

B) $E = \frac{(0.8\pi\varepsilon_0)(1cm)}{4\pi\varepsilon_0 (2m)^3}\left(2\cos(90^\circ)\hat{R} + \sin(90^\circ)\hat{\theta}\right)$

$= 0.25\,\hat{\theta}\,\frac{mV}{m}$

4.38

$$\vec{E} = \frac{18}{R^2}\hat{R}$$

$$V = -\int \vec{E}\, d\vec{l}$$

$$V_{AB} = -\int_{-4}^{2} E\, dz$$

$$= -\int_{-4}^{2} \left(\frac{18}{z^2}\right) dz$$

$$= -18 \int \frac{1}{z^2}\, dz$$

$$= -18 \cdot \left[-\frac{1}{z}\right]_{-4}^{2}$$

$$18 \cdot \frac{3}{4}$$

$$V_{AB} = 13.5\,V$$

(4.39)

$$V = \frac{\rho_l}{2\pi\varepsilon_0} \ln\left(\frac{r_2}{r_1}\right)$$

$$r_1 = \sqrt{(3-2)^2 + 4^2} = \sqrt{17}\ m$$

$$r_2 = 2\ m$$

$$V_{AB} = \frac{9\times10^{-9}}{2\pi \times 8.86\times10^{-12}} \ln\left(\frac{2}{\sqrt{17}}\right) = -117.09\ V$$